

3.3V/2A
 $F_{sw}=1.5\text{MHz}$
 $L=1\mu\text{H}$
 $I_{c(RMS1)} = 0.66\text{A}$

$D = V_{out}/V_{in} = 3.3/5 = 0.66$
 $I_{Ripple} = (V_{out}(1 - V_{out}/V_{in})) / (L \cdot f)$
 $I_{In} = I_{out} \cdot (V_{out}/V_{in})$
 $I_{c(RMS)} = I_{out} \cdot \text{SQRT}[(V_{out}/V_{in})(1 - (V_{out}/V_{in}))]$
 $I_{c(RMS)tot} = \text{SQRT}[I_{c(RMS1)}^2 + I_{c(RMS2)}^2]$

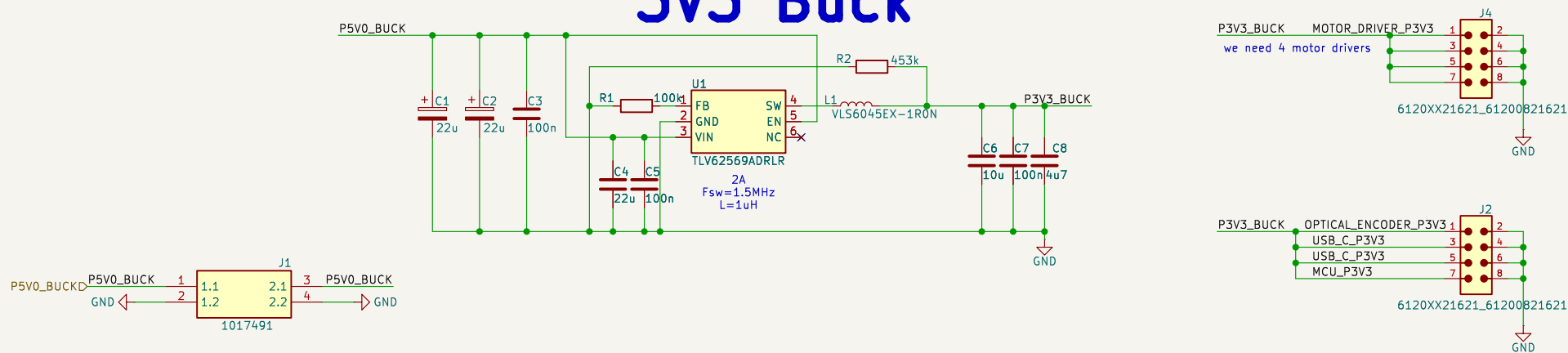
$I_{Ripple} = 0.748\text{A}$
 $I_{In} = 1.32\text{A}$
 $I_{c(RMS2)} = 0.947\text{A}$
 $I_{c(RMS)tot} = 1.155\text{A} \rightarrow 1.4\text{A}$

1, find cap bank > 1.4A rms ripple current

2, $Ripple_V_{out} = I_{Ripple} \cdot ESR$
 $33\text{mA} = 0.748\text{A} \cdot ESR, ESR=44.1\text{m}\Omega$

3, 1.32A is drawn from the previous 5V stage

3V3 Buck



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